

be verified directly from the corresponding properties for real numbers.

Algebraic Properties of $\mathbb{R}^n$

For all u, v, w in $\mathbb{R}^n$ and all scalars c and d :

- | | |
|--|-----------------------|
| (i) $u+v = v+u$ | (v) $c(u+v) = cu+cv$ |
| (ii) $(u+v)+w = u+(v+w)$ | (vi) $(cd)u = c(du)$ |
| (iii) $u+0 = 0+u = u$ | (vii) $c(du) = (cd)u$ |
| (iv) $u+(-u) = -u+u = 0$,
where $-u$ denotes $(-1)u$ | (viii) $1u = u$ |

For simplicity of notation, a vector such as $u + (-1)v$ is often written as $u - v$. Figure 7 shows $u - v$ as the sum of u and $-v$.

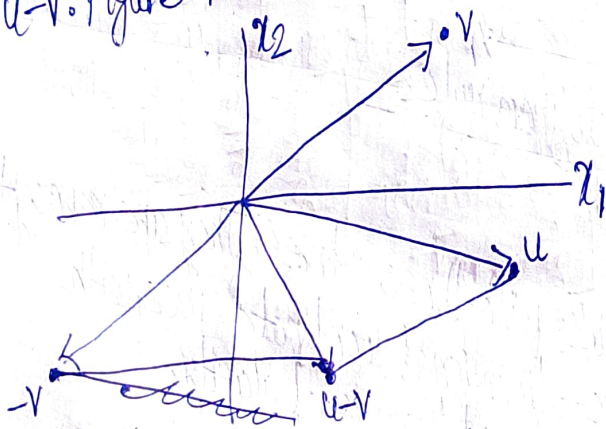


Figure 7: Vector Subtraction

Linear Combinations

Given vectors $v_1, v_2, \dots, v_p$ in $\mathbb{R}^n$ and given scalars $c_1, c_2, \dots, c_p$, the vector y defined by $y = c_1 v_1 + \dots + c_p v_p$ is called a linear combination of $v_1, \dots, v_p$ with weights $c_1, \dots, c_p$. Property (ii) above permits us to omit parentheses when forming such a linear combination. The weights in a linear combination can be any real numbers, including zero. For ex, some linear combinations of vectors v_1 and v_2 are $\sqrt{3}v_1 + v_2$, $\frac{1}{2}v_1$ ($= \frac{1}{2}v_1 + 0v_2$), and $0 = (0v_1 + 0v_2)$.